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A swin transformer-driven framework for gesture recognition to assist hearing impaired people by integrating deep learning with secretary bird optimization algorithm

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ABSTRACT

Hand gestures (HG) are the key communication technique for hearing-impaired people, which poses a problem for millions of individuals globally after communicating with those who don't have hearing impairments. The importance of technology in improving accessibility and thus raising the standard of living for persons with hearing impairments is globally acclaimed. Machine learning (ML) is a section of artificial intelligence (AI) that concentrates on developing a method that depends on data. The major problem of HG recognition is that the machine does not identify the human language straightforwardly, and human–machine interaction is required of media for communication, which is determined by machines and, in addition to humans, to assist hearing-impaired individuals and ageing people. Thus, HG recognition as a communication media is necessary to provide instructions to the computer. This paper proposes the Swin Transformer-Driven Framework for Gesture Recognition by Integrating Deep Learning with the Secretary Bird Optimization (STFGR-IDLSBO) methodology. The main intention of the STFGR-IDLSBO methodology is to develop an efficient and robust system for gesture recognition to assist hearing-impaired persons. Initially, the proposed STFGR-IDLSBO method utilizes adaptive bilateral filtering (ABF) in the image pre-processing stage to reduce noise while preserving the edges of the gestures in the captured images. Furthermore, the swin transformer (ST) is a feature extractor that effectively captures multiscale representations and spatial hierarchies from gesture images. The hybrid model integrates the convolutional neural network and bi-directional long short-term memory (CNN-BiLSTM) technique, which is employed for the gesture classification process. Finally, the secretary bird optimizer algorithm (SBOA) is utilized for the optimum hyperparameter tuning of the CNN-BiLSTM classifier. To ensure the enhanced performance of the STFGR-IDLSBO methodology, a wide range simulation investigation is performed under the Traffic Police Gesture dataset. The performance validation of the STFGR-IDLSBO technique portrayed a superior accuracy value of 99.25% over existing methods.

1. Introduction

In concordance with data from the World Federation of the Hearing Impaired and the World-Health-Organization are expected that 72 million individuals globally struggle with hearing muteness, comprising 360 million of the hearing-impaired population, with 32 million amongst other children [1]. Many of the speech and hearing-impaired populace face challenges in traditional literacy. For that, communication between mute and hearing-impaired people generally happens through gestures like sign language (SL), which depends on encompassing finger shapes, HG, manual expressions, and facial expressions to transfer meaning [2]. However, despite its vital role in bridging communication gaps, SL does have its restrictions. It contains the requirement for limited vocabulary, broad hand movements, and the

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difficulties of mastering this communication form [3]. Linguists have concluded that spoken languages and SL share several aspects. Like spoken languages, which utilize sound units to create words, SL utilizes form units. A gesture is a physical hand movement, arms, fingers, and other human body parts that permits individuals to communicate information and meaning with each other [4]. The vision-based and data gloves methods are dual alternate methods for HCI. The classification and detection of hand motions were analyzed using the vision-based method [5]. The most logical approach to generate an adaptable and convenient interface among gadgets and consumers is to utilize HGs. HCI methods can use gaming, gesture recognition, and virtual object manipulation applications. Hand tracking is a hypothetical field of computer vision (CV) that deals with three main components: hand tracking, hand segmentation, and hand part detection. HGs are the optimum interconnecting method and the most common notion in a gesture detection method [6].

HGs might be recognized utilizing most succeeding approaches: gesture is dynamic hand motion with or without movement, and posture is static hand-form ratio without movement. Any camera might identify any HG method; nevertheless, diverse cameras have different resolution assets. Recent research in deep learning (DL) models has shown significant advances in the field of SL recognition (SLR), depicting favourable results [7]. Novel structures utilizing DL frameworks show possibilities for attaining signer-independent SLR. Multiple reviews and surveys have initiated the task of delineating the obstacles, technical approaches, and probable avenues for investigation in SLR. These inclusive assessments underscore the crucial role of classification models, DL methods, and CNN-based approaches to advancing the area. Emphasis is also located on the importance of datasets and the SL gesture classifications, explaining their vital role in assisting growth [8]. Additionally, the preview of SLR enlarges overall applications to comprise particular fields like hand pose and fingerspelling detections. These domain-specific developments cater to the dissimilar linguistic necessities inherent to various SLs. The growing number of individuals with hearing impairments accentuates the requirement for more effectual communication methods, as conventional communication systems may not fully address their requirements [9]. While a crucial tool, SL presents challenges due to its reliance on physical gestures, making it difficult for many to learn and use. There is a robust demand for advanced technologies that can bridge the communication gap by improving the ease and accuracy of gesture-based interactions. Developing a more effective gesture recognition system could significantly improve the quality of life for those affected, enabling smoother communication and fostering greater inclusion. This motivates the exploration of innovative solutions like DL and optimization methods to improve gesture recognition for hearing-impaired individuals [10].

This paper proposes the Swin Transformer-Driven Framework for Gesture Recognition by Integrating Deep Learning with the Secretary Bird Optimization (STFGR-IDLSBO) methodology. The main intention of the STFGR-IDLSBO methodology is to develop an efficient and robust system for gesture recognition to assist hearing-impaired persons. Initially, the proposed STFGR-IDLSBO method utilizes adaptive bilateral filtering (ABF) in the image pre-processing stage to reduce noise while preserving the edges of the gestures in the captured images. Furthermore, the swin transformer (ST) is utilized as a feature extractor to effectively capture multiscale representations and spatial hierarchies from gesture images. The hybrid model integrates the convolutional neural network and bi-directional long short-term memory (CNN-BiLSTM) technique, which is employed for the gesture classification process. Finally, the secretary bird optimizer algorithm (SBOA) is utilized for the optimum hyperparameter tuning of the CNN-BiLSTM classifier. To ensure the enhanced performance of the STFGR-IDLSBO methodology, a wide range simulation investigation is performed under the Traffic Police Gesture dataset. The key contribution of the STFGR-IDLSBO methodology is listed below.

- The STFGR-IDLSBO model utilizes the ABF method for image pre-processing, which helps reduce noise while maintaining essential image features. This pre-processing step improves the clarity and quality of input data, which is significant for accurate gesture recognition. Preserving edges and details assists in more effective feature extraction in subsequent model stages.
- The STFGR-IDLSBO approach integrates an ST model as a feature extractor, taking advantage of its capability to capture multiscale, hierarchical features. This approach improves the model's ability to recognize intrinsic HG patterns. The ST enhances gesture recognition accuracy in diverse scenarios by effectively processing spatial information.
- The STFGR-IDLSBO technique integrates CNN with BiLSTM models for classification, combining spatial and temporal feature learning. This hybrid methodology enables effectual sequence modelling, improving the recognition of dynamic gestures. By capturing both spatial patterns and sequential dependencies, it crucially enhances the accuracy of gesture classification.
- The STFGR-IDLSBO method employs the SBOA model to fine-tune hyperparameters, ensuring optimal performance in gesture recognition tasks. This optimization process improves the efficiency and accuracy of the model by adjusting key parameters for better generalization. The unique SBOA approach assists in achieving faster convergence and enhanced recognition outputs.
- This method tackles spatial and temporal complexities in gesture recognition by incorporating ST with a CNN-BiLSTM hybrid classification model optimized through SBOA. ST allows for effectual multiscale feature extraction, while the CNN-BiLSTM model captures spatial patterns and sequential dependencies. SBOA fine-tunes hyperparameters to maximize performance, giving a novel and highly efficient solution for accurate, real-time gesture recognition.

2. Related works

Onaizah [11] develops a Robust Gesture SL Recognition Utilizing Chicken Earthworm Optimization with DL (RSLR-CEWODL) method. The projected RSLR-CEWODL model employs a ResNet-101 technique for feature extraction. Also, the RSLR-CEWODL model utilizes a Whale Optimizer Algorithm (WOA) with a Deep Belief Network (DBN) approach for the SLR approach. Kumar et al. [12] developed a novel smart application to improve communication for people with disabilities. This application harnesses progressive technologies like CV, gesture recognition, and AI by utilizing real-world SL interpretation and detection. Its primary objective is overcoming communication difficulties and facilitating smooth conversations between non-SL and SL users. Hameed et al. [13] introduced an RFID-based smart mask for a lip-reading structure that can read lips under face masks, allowing effectual speech recognition and fostering conversational availability for people with hearing impairment. Alrowais et al. [14] introduced an innovative Bayesian Optimizer with DL-driven HGR Based SL Communication (BODL-HGRSLC) for Disabled Individuals. The developed BODL-HGRSLC model incorporates the concepts of DL models CV. In the projected model, a DCNN-based ResNet technique is utilized for the feature extractor. Also, the projected method utilizes BO for the hyperparameter tuning process. Finally, a bi-directional GRU (BiGRU) technique is used for the HGR process. Fatima et al. [15] introduced a video-based ASL identification method. This method employs OpenCV for image normalization, MediaPipe for hand tracking, and Gesture Control CNN for gesture localization. Applied in Python, it filters hand regions, records video streams, and identifies ASL letter gestures with greater precision. Utilizing ML and CV, this method breaks communication barriers, promotes inclusivity, improves user experience, and aids available technologies.

Sukruth et al. [16] introduced a sensor-based HGR wearable gadget. The presented approach includes designing the hand glove by incorporating the Inertial Measurement Unit (IMU) and flex sensors in the HGR

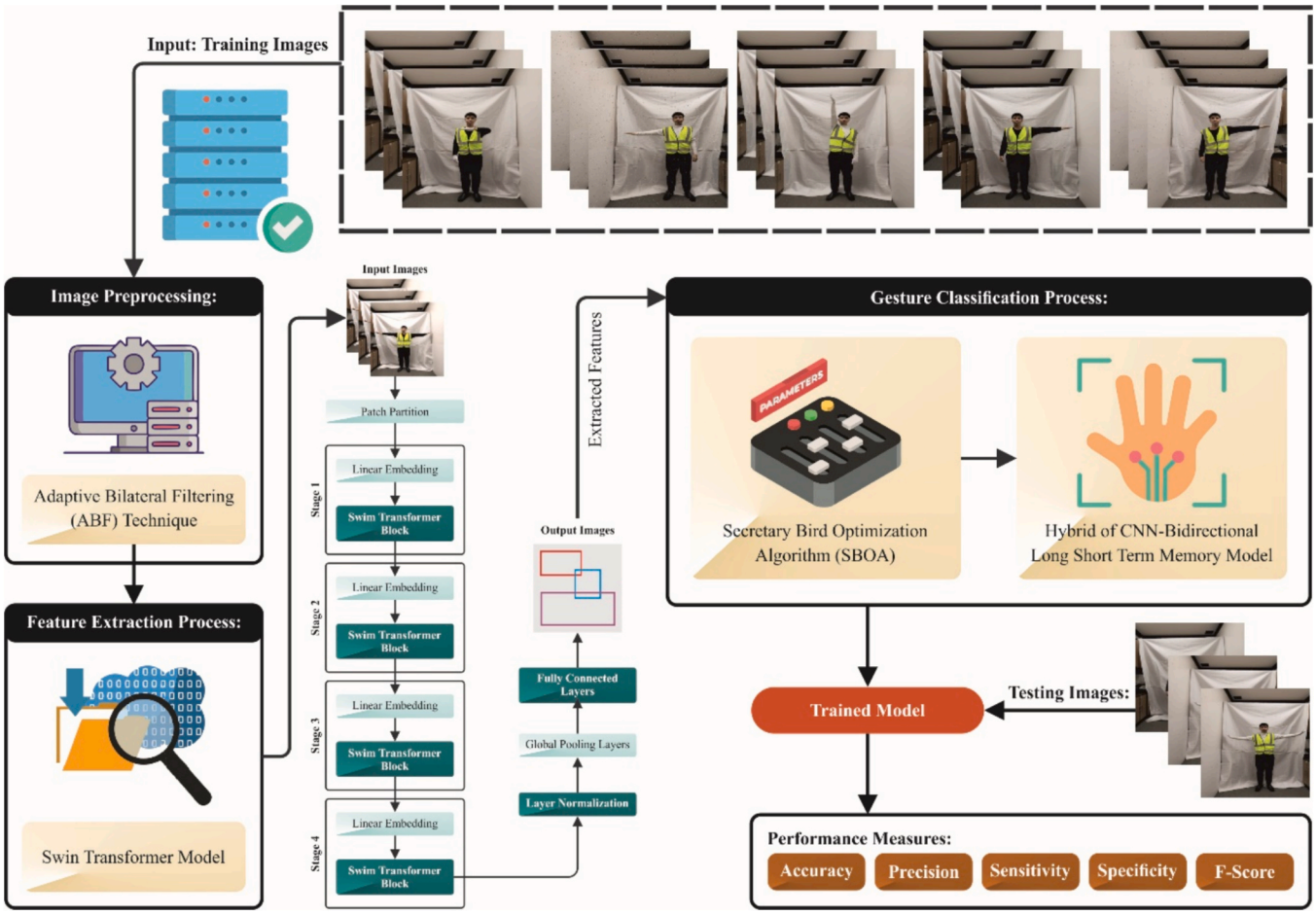


Fig. 1. Workflow of STFGR-IDLSBO model.

wearable gadgets. They are depicted by the ISL dictionary utilizing ML models. The study aims to improve the HGR ability by presenting the Dynamic Spatio-temporal Warping (DSTW) model and LSTM-based learning techniques. Rangdale et al. [17] introduce a Programmed interpretation structure for manual letter token sequence in English communication through gestures. It handles uncovered hand pictures that allow the client to cooperate with the structure. The presented DL technique utilizing CNN with the ISL (Indian SL) Dataset is applied. This technique contains pre-processing the dataset to remove features from the image and then training CNN to identify the gestures and signs with audio. In [18], a developed flashfly optimization-based ANN is introduced for effectual HG identification, and to overwhelm the present challenges, to create the flashfly optimization-based ANN model is presented. The primary goal of the optimization approach is to effectually tune the parameters comprised in ANN by achieving an optimum solution. Maashi, Iskandar, and Rizwanullah [19] introduce the Smart Assistive Communication System for the Hearing-Impaired using SL Recognition with Hybrid Deep Learning (SACHI-SLRHDL) technique using SL recognition with IoT. It enhances image quality with bilateral filtering, extracts features using MobileNetV3 and classifies with a CNN-BiGRU-A model. The Attraction-Repulsion Optimization Algorithm (AROA) model fine-tunes the model's hyperparameters for better performance. El-Kenawy et al. [20] propose the Greylag Goose Optimization (GGO) approach, inspired by the migratory behaviour of geese, as a new swarm-based optimizer. Geese fly in a "V" formation to reduce air resistance and travel farther, which the algorithm emulates.

Kadhim, Der, and Phing [21] recognize dynamic HGs for individuals with finger disabilities by segmenting gestures with Otsu's algorithm and utilizing motion data. The model incorporates a CNN with a recurrent neural network (RNN) by employing Inception-v3 for feature extraction and LSTM for classification. Baihan et al. [22] propose CNNSa-LSTM, a model incorporating CNN, Self-Attention, and LSTM for SL recognition. The method extracts spatial features with VGG16 and motion features with optical flow. A hybrid optimizer (HO) technique using the Hippopotamus and Pathfinder models improves performance in SL tasks. Goel, Bansal, and Gupta [23] propose the autoencoders and adaptive grey wolf optimization (AEGWO-Net) technique by integrating ML and swarm intelligence. It extracts features using HOG, reduces dimensions with an autoencoder, and refines features with an improved GWO model. Mahmoud, Ziedan, and Zamel [24] present a novel approach for recognizing Arabic sign alphabets by integrating CNN with five ML approaches. The method evaluates the model and uses six optimization techniques to improve classification accuracy. Gaikwad and Shete [25] present a deep model for HG recognition in ISL. The method utilizes image collection, segmentation, and edge detection with ATRG-CED, optimized by the Modified Tasmanian Devil Optimization (MTDO) method for improved performance. Subhashini and Revathi [26] introduce a Gabor Line Derivative-based feature extraction technique integrated with a Levy flight Whale optimization-driven Deep CNN (DCNN) model for recognizing static and dynamic HG images. Hariprasath et al. [27] developed wearable flexible microsensors for motion-based healthcare applications, overcoming issues like bulk and rigidity. These sensors, utilized in devices such as sensing gloves, capture detailed movement data for gestures, such as in Indian SL (ISL), by detecting changes in response to body motions. Noor et al. [28] propose a hybrid model to capture the spatio-temporal aspects of SL. The hybrid model comprises a CNN classifier for extracting spatial features from SL data and an LSTM classifier to extract spatial and temporal

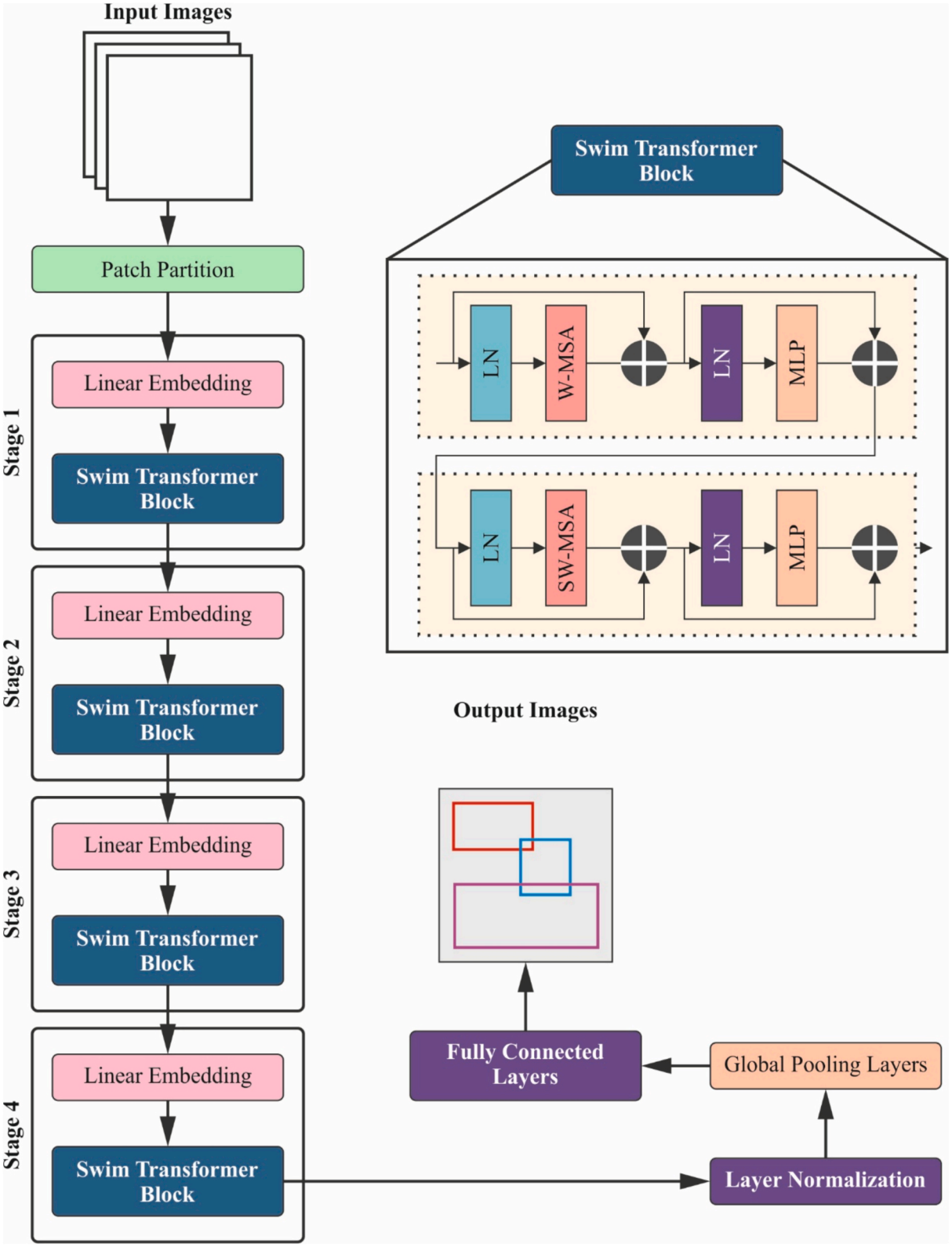


Fig. 2. Structure of ST model.

characteristics to handle sequential data.

Despite the improvements in SLR using DL and optimization techniques, various limitations remain. Many approaches depend heavily on large, well-labelled datasets, which may not always be available, particularly for diverse SLs. Moreover, the complexity of real-time gesture recognition in dynamic environments continues to pose challenges regarding accuracy and computational efficiency. Some methods also face difficulty handling lighting, background, and hand positioning discrepancies, affecting performance. Furthermore, although promising, wearable sensors face issues of bulkiness and rigidity, limiting their widespread adoption. The requirement for more effectual and flexible solutions for real-time, robust, and cross-language

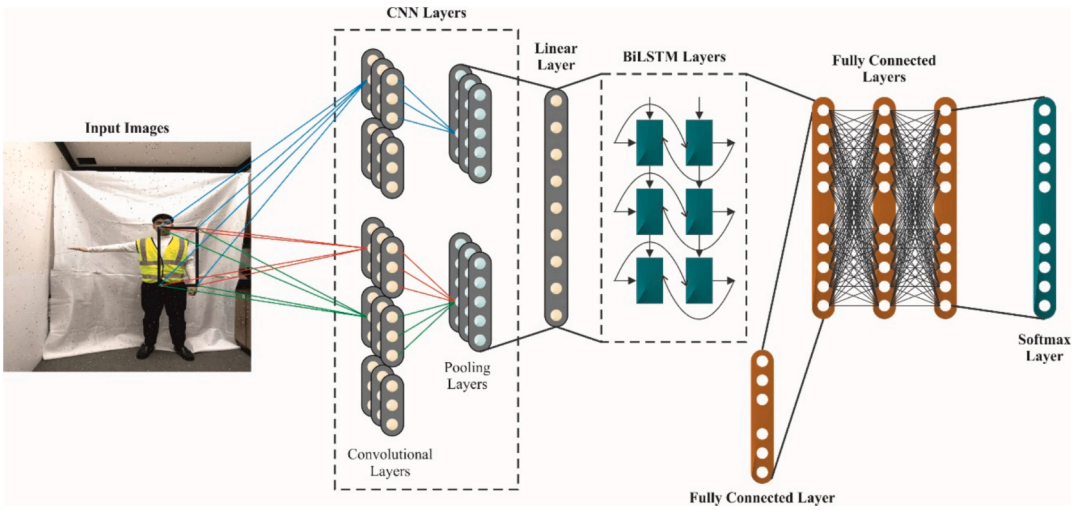


Fig. 3. Architecture of CNN-Bi-LSTM model.

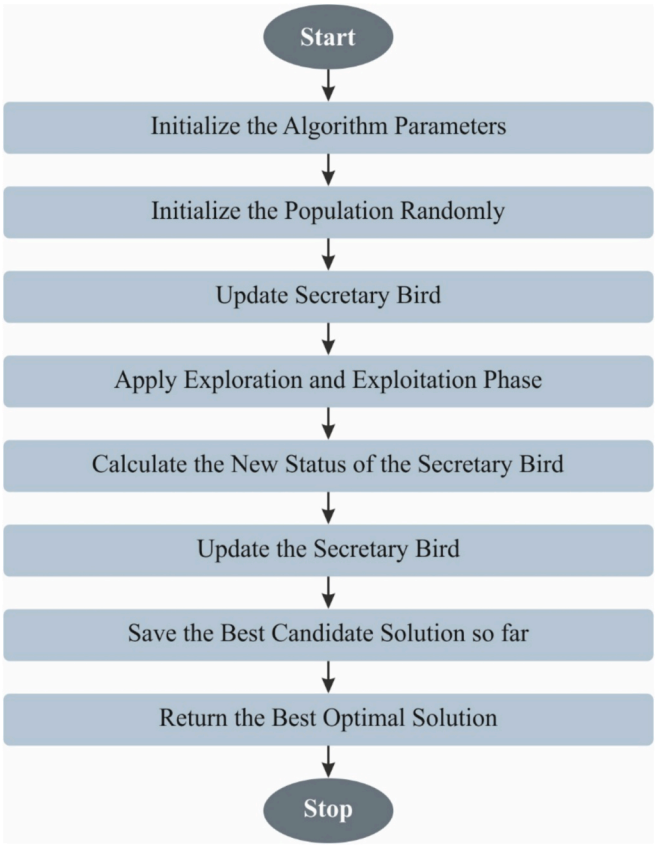


Fig. 4. Steps involved in the SBOA method.

recognition remains a key research gap. There is also a lack of standardized benchmarks to compare the efficiency of diverse models across several SLs.

3. Methodology

This paper proposes the STFGR-IDLSBO methodology. The main intention of the method is to develop an efficient and robust system for gesture recognition to assist hearing-impaired persons. It contains four stages: image pre-processing, ST for feature extraction, gesture classification using CNN-BiLSTM, and parameter tuning using SBOA. Fig. 1 illustrates the workflow of the STFGR-IDLSBO model.

Table 1
Details on Dataset.

Gesture	Image Count
Lane Left	50
Lane Right	50
Left	50
Left Over	50
Left Turn	50
Move Straight	50
Right	50
Right Over	50
Right Turn	50
Stop Signal	50
Total Image	**500**

3.1. Stage I: Image pre-processing

Initially, the proposed STFGR-IDLSBO method utilizes ABF in the image pre-processing stage to reduce noise while preserving the edges of the gestures in the captured images [29]. This model is chosen because it can reduce noise while preserving crucial edges and fine details, which is significant for accurate gesture recognition. Unlike conventional filters that can blur edges, ABF effectually handles both noise reduction and edge preservation, making it ideal for processing gesture images where sharp boundaries are significant. Furthermore, ABF adjusts the strength of the filter based on local image characteristics, ensuring optimal processing for different areas of the image. This adaptability makes it more effectual than static filtering techniques, improving performance in dynamic gesture recognition tasks where varying backgrounds and noise levels are common.

ABF is an effective image pre-processing approach applied in gesture recognition methods for hearing-impaired people, targeted at improving image quality while maintaining essential edge details. This model uses intensity differences or spatial proximity to smooth the image, decreasing noise without blurring important gesture characteristics. Adjusting to local image features guarantees that the most related characteristics of HGs remain sharper and different, which is essential for precise recognition. In terms of hearing-impaired communication, such filtering improves the sturdiness of gesture detection, enhancing system performance in varied atmospheres. Using ABF outcomes in precise input data maintains more accurate classification and feature



Fig. 5. Sample images.

extraction in DL methods.

3.2. Stage II: ST model

Next, the ST is utilized for feature extraction to effectively capture multiscale representations and spatial hierarchies from gesture images [30]. This model is chosen due to its capability to capture hierarchical, multiscale features, which is significant for comprehending intrinsic gesture patterns in SLR. Unlike conventional CNNs, the ST utilizes a shift-window mechanism, enabling enhanced modelling of long-range dependencies while maintaining computational efficiency. This feature is specifically beneficial in gesture recognition, where spatial context

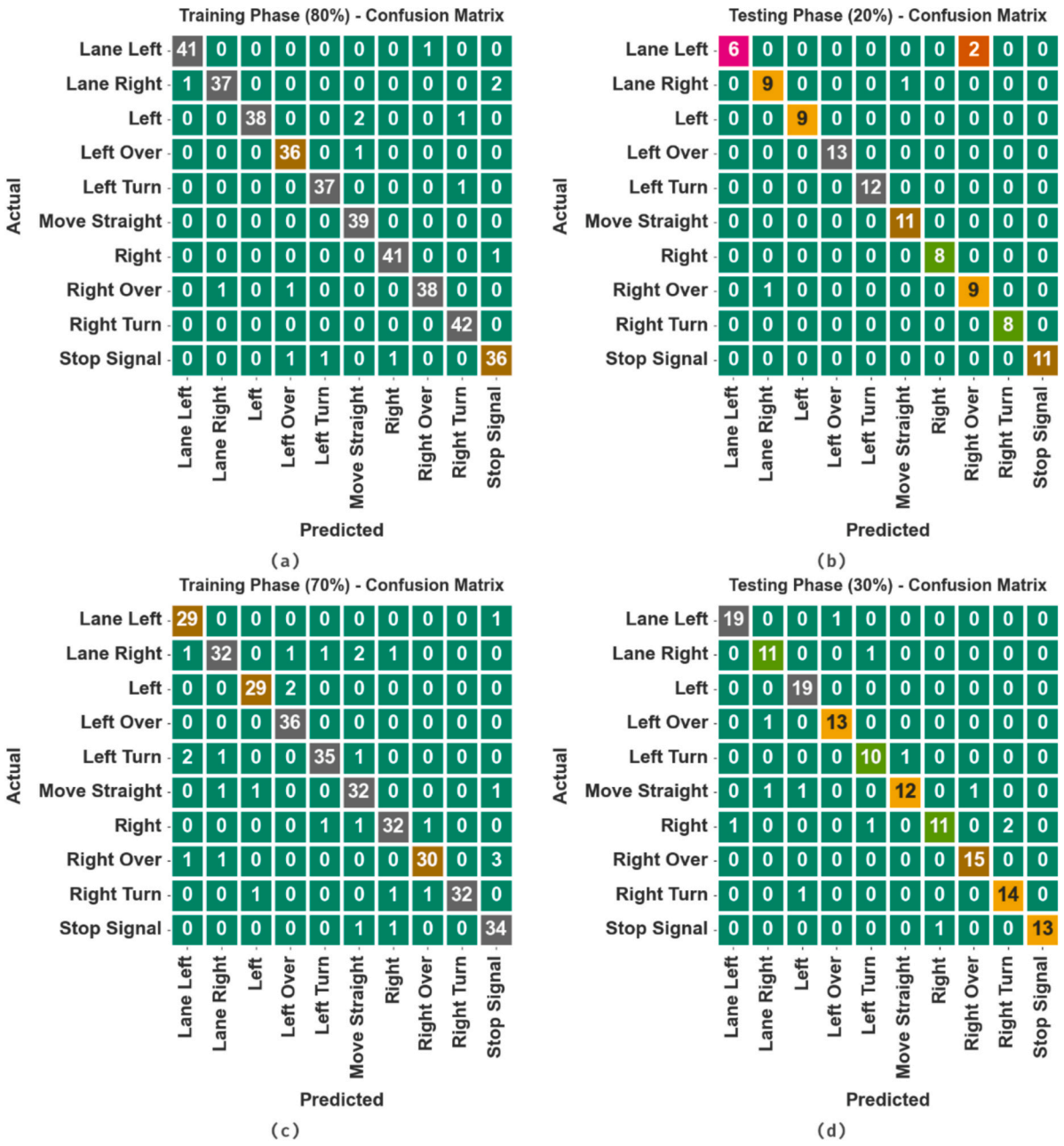


Fig. 6. Confusion matrix of (a-b) 80% and 20% of TRASE/TESSE and (c-d) 70% and 30% of TRASE/TESSE.

and global relationships between parts of the hand or body must be considered. Moreover, STs adapt to varying input sizes and resolutions, making them highly versatile. Their superior performance in capturing local and global features makes them an ideal choice over conventional methods, which might face difficulty with complex, large-scale input data. Fig. 2 illustrates the ST framework.

The ST utilizes a hierarchic model for feature extraction, permitting dissimilar layers to capture different receptive areas while calculating self-attention (SA). This allows for the improved usage of global information and longer-range semantic relations in hand-deep images. Hence, they apply the ST as a backbone to make feature mapping from multiple perspectives simulated by the input.

Mainly, the input adaptable multiple perspective representations signified as $x \in \mathbb{R}^{H\times W\times C}$, are separated into n non-overlapping patches that are successively predictable onto $1D$ vectors. It is strengthened by position embedding, which is further handled by ST blocks. It is built by substituting the normal multi-head SA (MSA) component in a conventional block of Transformer using a shifted-window-based attention mechanism. During this application, the size of the window is fixed to 11. All ST blocks were computed as shown:

$$\begin{cases} w^{f} = MLP\left[LN\left(w^{f}\right)\right] + w^{f} \\ \widehat{w}^{\mathrm{f}} = W - MSA\left[LN\left(w^{f-1}\right)\right] + w^{f-1} \\ w^{f+1} = MLP\left[LN\left(w^{f+1}\right)\right] + w^{f+1'} \\ \widehat{w}^{f+1} = \mathrm{SW} - MSA\left[LN\left(w^{f}\right)\right] + w^{f} \end{cases} \tag{1}$$

Meanwhile, $W-MSA$ and $SW-MSA$ individually characterize the window-based MSA and the shifted-window separating configurations in the ST. W^{f} and w^{f} correspondingly characterize the vectors of the output patch from the $(S)W\text{-}MSA$ and MLP units of the block of the transformer f. LN represents layer normalization, and MLP represents numerous fully connected (FC) layers.

The Transformer blocks are separated into N phases over the layers of patch merging. The lower-layer and output characteristics from the last

Table 2
Gesture detection of STFGR-IDLSBO technique with 80%TRASE and 20%TESSE.

Class Labels	$Accu_y$	$Prec_n$	$Sens_y$	$Spec_y$	F_{Score}
TRASE (80 %)					
Lane Left	99.50	97.62	97.62	99.72	97.62
Lane Right	99.00	97.37	92.50	99.72	94.87
Left	99.25	100.00	92.68	100.00	96.20
Left Over	99.25	94.74	97.30	99.45	96.00
Left Turn	99.50	97.37	97.37	99.72	97.37
Move Straight	99.25	92.86	100.00	99.17	96.30
Right	99.50	97.62	97.62	99.72	97.62
Right Over	99.25	97.44	95.00	99.72	96.20
Right Turn	99.50	95.45	100.00	99.44	97.67
Stop Signal	98.50	92.31	92.31	99.17	92.31
Average	**99.25**	**96.28**	**96.24**	**99.58**	**96.22**
TESSE (20 %)					
Lane Left	98.00	100.00	75.00	100.00	85.71
Lane Right	98.00	90.00	90.00	98.89	90.00
Left	100.00	100.00	100.00	100.00	100.00
Left Over	100.00	100.00	100.00	100.00	100.00
Left Turn	100.00	100.00	100.00	100.00	100.00
Move Straight	99.00	91.67	100.00	98.88	95.65
Right	100.00	100.00	100.00	100.00	100.00
Right Over	97.00	81.82	90.00	97.78	85.71
Right Turn	100.00	100.00	100.00	100.00	100.00
Stop Signal	100.00	100.00	100.00	100.00	100.00
Average	**99.20**	**96.35**	**95.50**	**99.55**	**95.71**

phase are then connected to the HFE units to estimate the pose for all views. Compared with the raw ST, 4x*and* 8 x downsampling and forego 16 x downsampling are implemented to preserve fine spatial information and decrease the dimension of the model.

3.3. Stage III: Gesture classification using CNN-BiLSTM

For the gesture classification process, the hybrid model integrating the CNN-BiLSTM model is utilized [31]. This hybrid model is chosen because it can effectually capture spatial and temporal features, making it ideal for recognizing dynamic gestures. The CNN outperforms in extracting spatial features from images and learning key visual patterns such as hand shapes and movements. The BiLSTM then captures the temporal dependencies, learning the sequence of gestures over time. This integration allows the model to comprehend the context and flow of gestures, which is crucial in SLR. Compared to single models like CNN or LSTM, this hybrid approach gives a more comprehensive understanding of gesture dynamics, significantly enhancing classification accuracy in complex gesture sequences. Fig. 3 demonstrates the architecture of CNN-Bi-LSTM.

This section describes the hybrid CNN-BiLSTM, which incorporates the powers of CNN and BiLSTM for removing valued information from time-series data. Noise removal is attained over dimensionality reduction, guaranteeing that just related information rests in the diminished matrix. Useful information is then exposed over the convolution process, whereas the kernel matrix or filters emphasize dissimilar features. This framework utilizes the strength of a BiLSTM approach to predict shorter and longer-term time-series data. The basic to effective training of the BiLSTM approach is its capability to simultaneously handle input data in forward and backwards directions.

Table 3
Gesture detection of STFGR-IDLSBO technique with 70%TRASE and 30%TESSE.

Class Labels	$Accu_y$	$Prec_n$	$Sens_y$	$Spec_y$	F_{Score}
TRASE (70 %)					
Lane Left	98.57	87.88	96.67	98.75	92.06
Lane Right	97.43	91.43	84.21	99.04	87.67
Left	98.86	93.55	93.55	99.37	93.55
Left Over	99.14	92.31	100.00	99.04	96.00
Left Turn	98.29	94.59	89.74	99.36	92.11
Move Straight	97.71	86.49	91.43	98.41	88.89
Right	98.29	91.43	91.43	99.05	91.43
Right Over	98.00	93.75	85.71	99.37	89.55
Right Turn	99.14	100.00	91.43	100.00	95.52
Stop Signal	98.00	87.18	94.44	98.41	90.67
Average	**98.34**	**91.86**	**91.86**	**99.08**	**91.74**
TESSE (30 %)					
Lane Left	98.67	95.00	95.00	99.23	95.00
Lane Right	98.00	84.62	91.67	98.55	88.00
Left	98.67	90.48	100.00	98.47	95.00
Left Over	98.67	92.86	92.86	99.26	92.86
Left Turn	98.00	83.33	90.91	98.56	86.96
Move Straight	97.33	92.31	80.00	99.26	85.71
Right	96.67	91.67	73.33	99.26	81.48
Right Over	99.33	93.75	100.00	99.26	96.77
Right Turn	98.00	87.50	93.33	98.52	90.32
Stop Signal	99.33	100.00	92.86	100.00	96.30
Average	**98.27**	**91.15**	**91.00**	**99.04**	**90.84**

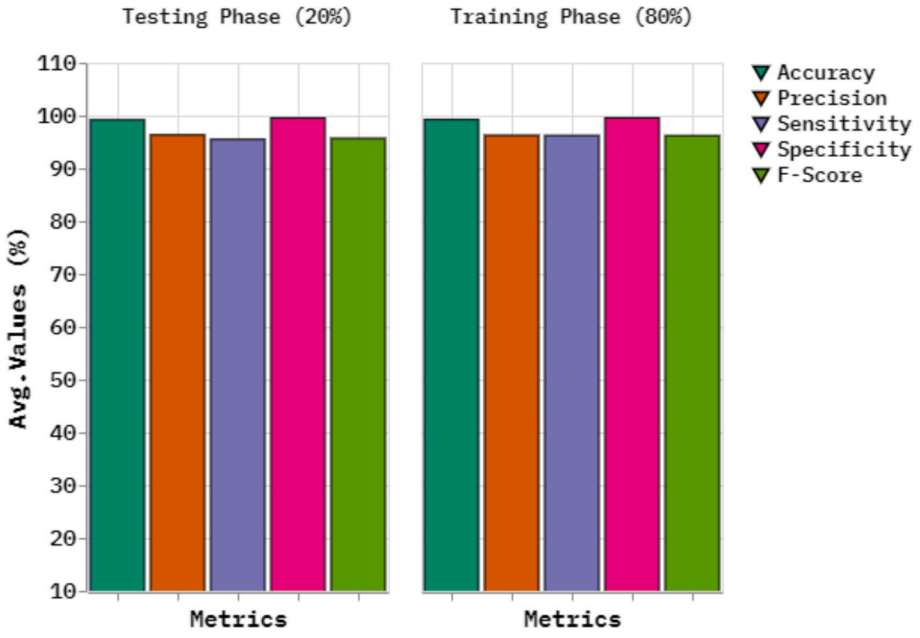


Fig. 7. Average of STFGR-IDLSBO technique with 80%TRASE and 20%TESSE.

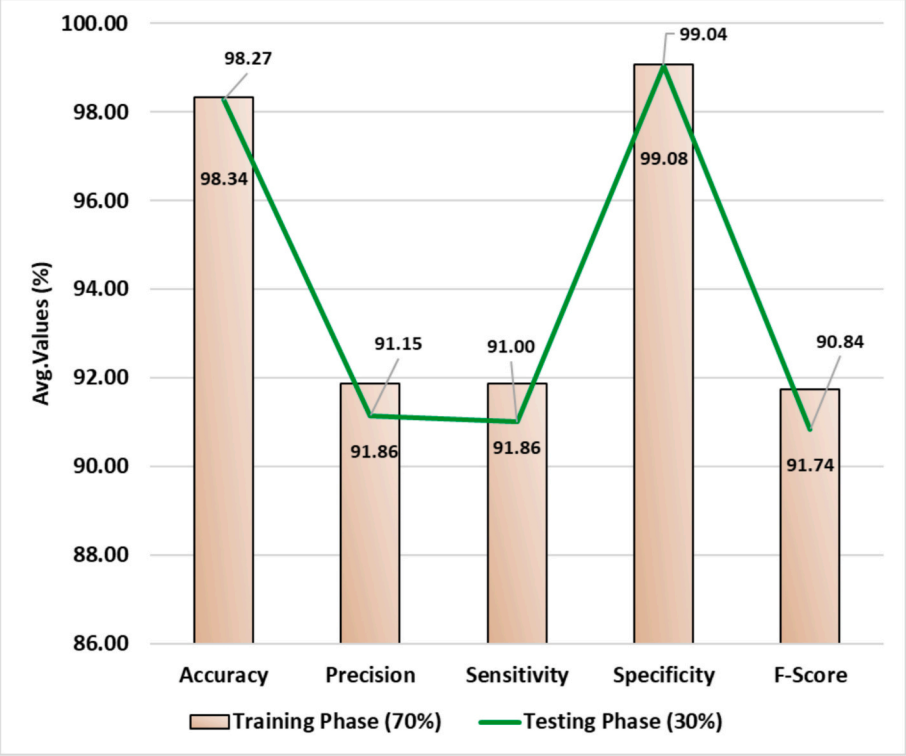


Fig. 8. Average of STFGR-IDLSBO technique with 70%TRASE and 30%TESSE.

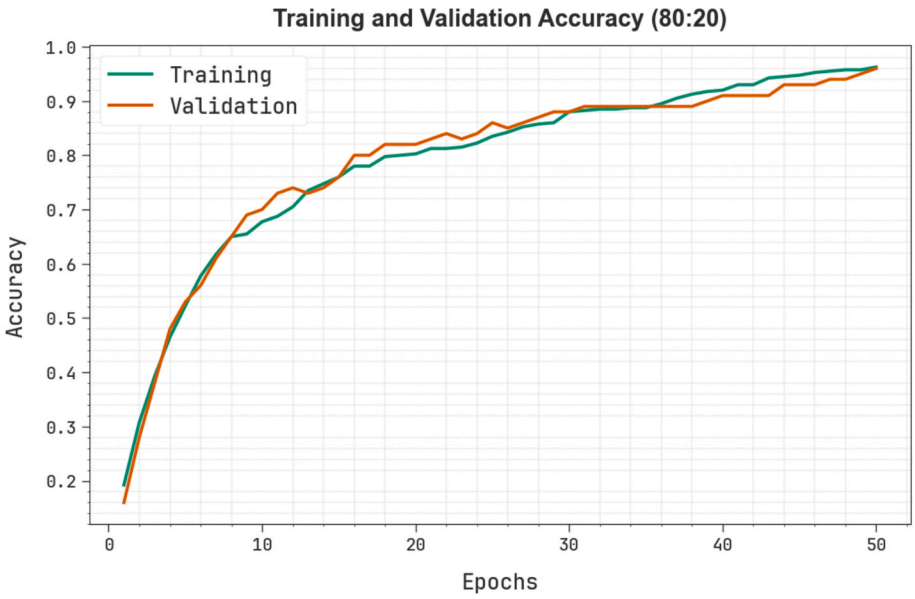


Fig. 9. $Accu_y$Curve of the stfgr-idlsbo technique with 80%trase and 20%tesse.

The outcomes of this approach are a mixture of contextual descriptions, giving a deeper understanding of the data context than a one-way LSTM approach. The CNN is responsible for encoding input data sequences by recognizing novel characteristics. In contrast, the BiLSTM decodes and understands temporal relations in the data stream to predict traffic designs. This approach is separated into two different portions −the CNN portion, with a CNN, flatten, and pooling layer, and the BiLSTM portion, which includes a BiLSTM layer following the dropout layer and the formerly dense layer.

CNN block. The primary aim of pooling and convolutional layers is typically examining data to remove essential characteristics. The convolutions are performed in the convolution layer, which collects nodes that join lower in addition to the upper layer edges and nodes by joining the input data with smaller matrices recognized as filters or kernels. All filters perform a convolutional process on parts of the matrix it cooperates with, making a convolution matrix showing a particular feature of the normal input matrix. Applying numerous filters results in a different variety of individual convoluted characteristics. An advanced method with a complex feature range will improve the input data signal compared to a process that depends only on the novel input data. Therefore, the algorithm's efficiency is enhanced by combining added filters. The convolution procedure is defined as shown: *I* refers to the

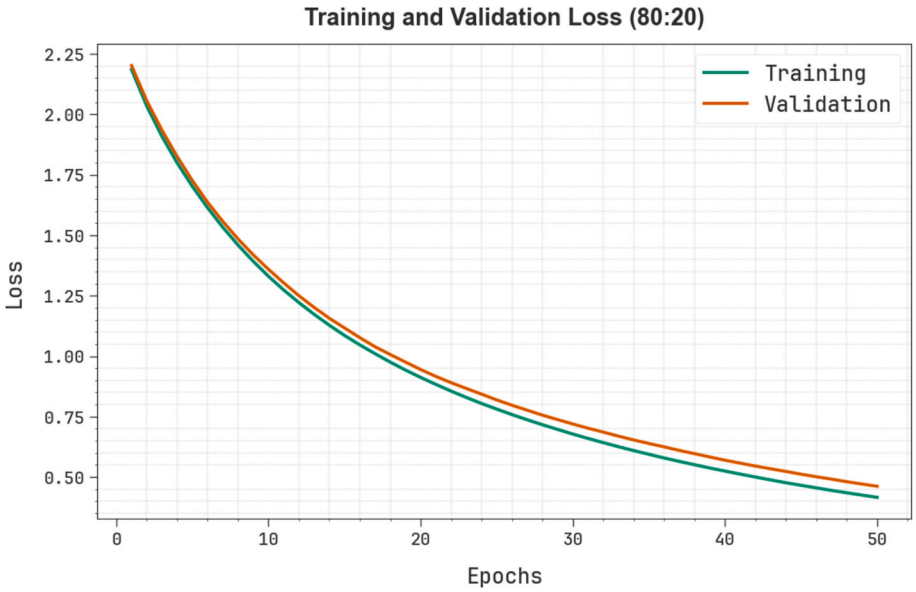


Fig. 10. Loss analysis of the STFGR-IDLSBO technique with 80%TRASE and 20%TESSE.

Table 4
Comparative outcomes of the STFGR-IDLSBO approach with other methodologies.

Method	$Accu_y$	$Prec_n$	$Sens_y$	$Spec_y$	F_{Score}
STFGR-IDLSBO	99.25	96.28	96.24	99.58	96.22
Fuzzy LSTM	93.12	94.74	92.08	92.61	92.92
CapsNets/ARPO	96.66	92.47	94.59	97.52	96.00
LSTM- DGCN	92.85	93.56	91.96	92.55	95.67
TPGR-SEMCN	98.95	92.61	93.01	94.61	92.67
LSTR-S3DRN	92.21	94.67	94.88	95.04	95.57
RFs Method	93.36	92.15	93.88	96.70	92.14
RNN	94.49	95.85	93.93	95.08	94.45
Linear SVM	92.46	94.20	91.35	92.07	92.15
ST-CNN	96.12	91.71	94.05	97.02	95.84
DT	92.10	92.87	91.20	91.96	95.13
ResNeXt-101	92.95	95.41	95.42	95.77	96.12
mXception	93.89	92.68	94.44	97.35	92.80
3D-ShuffleNetv2	95.28	96.48	94.53	95.72	95.20

input matrix produced from the removed time-series data Ex_T, J refers to the kernel matrix, and $(n, 0)$ characterizes the column and row indexes of the resultant matrix (S).

$$S[n, o] = (I.J)[n, o] \tag{2}$$

$$S[n, o] = \sum_k \sum_l J[k, l].[n-k, n-l] \tag{3}$$

The convolution layer in CNN methods normally utilizes the *ReLU* activation function, which is well-known for its extensive usage and efficacy. *ReLU* guarantees that not every neuron is stimulated simultaneously by converting each negative value to zero, leading to critical computational benefits. Certainly, *ReLU* is 6 times quicker for training than substitutions, namely *tanh*. The *ReLU* description is as shown:

$$f(S) = S^+ = \max(0, S) \tag{4}$$

The max −pooling layer, which emulates the convolution layer, is responsible for shrinking the matrix's size by removing a few values while keeping the main features, which all filters have been identified. A window navigates through the matrix correspondingly as the filter. The input values of all patches in the matrix are inspected, and all patches' maximum value is chosen. Therefore, this particular layer makes novel matrices, which assist as condensed forms of the convolution matrices. Next, the pooling layer, a flattened user, has been used to restructure the input of the following layer by flattening the values beside the last dimensions. The extracted features from the convolution portion are designated Y, which is also connected with the BiLSTM portion.

BiLSTM block. The following portion of this model comprises a BiLSTM component, which contains a BiLSTM model, then a dropout layer, after that dense layer. Describing how an LSTM technique works before coming into a BiLSTM model is essential. An LSTM technique is an RNN that outshines conventional RNNs because of its particular layer architecture. Whereas traditional RNNs fight to maintain longer-term dependencies as the system develops, they can process shorter-term memory for sequence data analysis, namely time series, by utilizing cyclic links in their hidden layers (HLs). LSTMs deal with this problem by performing as a conveyor belt-like memory cell, warehousing important information through each unit. All LSTM components contain a memory cell and three gates, which control the information flow in the method by defining what must be retained and rejected. This feature is essential for LSTM to recognize longer-term dependencies and tackle the issue. The following share would explore the parts of the input, output, and forget gates. The forget gate utilizes a sigmoid function to handle the input (Y) and the preceding HL $(s-1)$. This function is labelled as f_t, making an esteemed range between (0-1), demonstrating the percentage of data for storage. Furthermore, the input gate (j) additionally handles information from the current input and preceding HL by using various sigmoid functions to transform the information into values ranging between (0-1). The advanced data is then exposed to the tanh function for network security, giving a value inside the interval of (−1 to 1). The result on which information to retain is completed according to the incorporation of the sigmoid function and the tanh that leads to D_t. Based on this information, the cell state is verified. Initially, the preceding state (D) is changed by the forget outcome. Then, the output of the input gate can consist of modifying the cell state using new significant values acknowledged by the system, resulting in the upgraded cell state D_t. The output gate then utilizes the sigmoid function to analyze the current input data and the preceding HL, and the tanh function to assess the novel cell state to establish the following HL value. The results of (p) are combined to create the revised HL (s_t). These upgraded values for the cell and HL should be accepted in the following time steps. The processes carried out inside an LSTM cell are correctly defined as shown:

$$f_t = \sigma\left(W_f.[s_{t-1}, Y_t] + b_f\right) \tag{5}$$

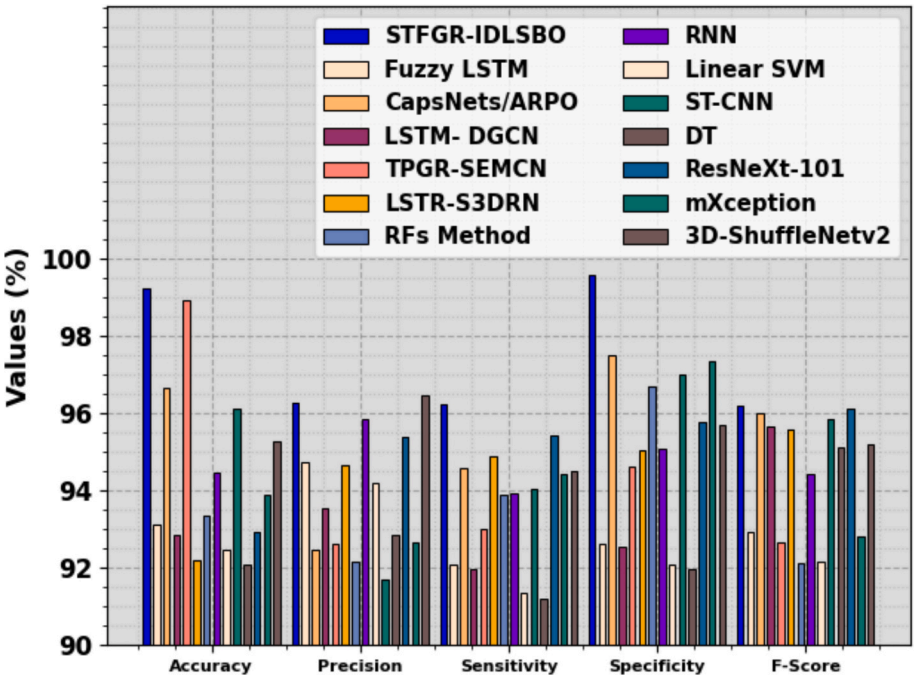


Fig. 11. Comparative analysis of STFGR-IDLSBO method with other approaches.

Table 5
TRAT results of STFGR-IDLSBO method with other techniques.

Method	TRAT (sec)
STFGR-IDLSBO	4.88
Fuzzy LSTM	8.05
CapsNets/ARPO	7.19
LSTM- DGCN	6.90
TPGR-SEMCN	9.87
LSTR-S3DRN	9.02
RF	9.13
RNN	6.94
Linear SVM	8.78
ST-CNN	9.08
DT	9.60
ResNeXt-101	10.03
mXception	8.98
3D-ShuffleNetv2	11.02

$$j_t = \sigma\left(W_j.[s_{t-1}, Y_t] + b_j\right) \tag{6}$$

$$\widetilde{D}_t = \tanh(W_D.[s_{t-1}, Y_t] + b_D) \tag{7}$$

$$D_t = f_t.D_{t-1} + j_t.\widetilde{D}_t \tag{8}$$

$$p_t = \sigma\left(W_p.[s_{t-1}, Y_t] + b_p\right) \tag{9}$$

$$s_t = D_t.\tanh(D_t) \tag{10}$$

Regarding the abovementioned equations, σ signifies a sigmoid function, and tanh has been applied to indicate the hyperbolic tangent function. y_t defines the input data and s_t compared to HL at *tth* time. W_f and W_y represent the biases and weights of input y. Later, when describing how an LSTM network works, the conversation moves on to BiLSTM techniques, which are constructed on LSTM techniques by combining Bi-directional RNNs. A forward and a backward LSTM method represents a BiLSTM model. The forward LSTM examines values from $t-l$ to t, whereas the backward LSTM inspects values from t to $t-l$.

Either the forward $\left(\overrightarrow{h}\right)$ or backward $\left(\overleftarrow{h}\right)$ systems give outputs in the same manner as an individual LSTM. Finally, the BiLSTM layer makes a variable Z_t utilizing a different equation.

$$Z_t = \sigma\left(\overrightarrow{h}, \overleftarrow{h}\right) \tag{11}$$

In this sample, a sigmoid function has been applied to combine the outcomes of the individual LSTM models. It is significant to recognize that Bi-LSTMs use dropout as a model to stop overfitting. The dropout rate is a vital hyperparameter to consider after describing the approach.

3.4. Stage IV: Parameter tuning using SBOA

Eventually, the SBOA is utilized for the optimum hyperparameter tuning of the CNN-BiLSTM classifier [32]. This technique is chosen because it can effectively explore the solution space and find optimal hyperparameters for complex models. SBOA is inspired by the migratory behaviour of birds, which allows it to balance exploration and exploitation efficiently, averting local minima and ensuring a more global search for optimal solutions. Unlike conventional optimization techniques, SBOA can adaptively adjust its search process, making it highly appropriate for tuning DL models with large and complex parameter spaces. Its robustness and flexibility allow it to outperform conventional optimization methods such as grid or random search, ensuring improved convergence and model performance. SBOA's efficiency in fine-tuning hyperparameters results in faster convergence and enhanced accuracy in tasks like gesture recognition. Fig. 4 depicts the steps involved in the SBOA methodology.

SBOA is a heuristic optimization model derived from the African Secretary Bird's hunting behaviour. Renowned for its hunting models and effective searching approaches, this model targets imitating these behaviours to solve complex optimizer problems. SBOA efficiently seeks the optimum search region solution by mimicking Secretary Bird's rapid running, ground searching, and accurate striking behaviours. The fundamental phases of SBOA are as demonstrated:

(1) Initialization stage of the algorithm: The locations of the bird group are primarily initialized by arbitrarily generating all locations of the secretary birds inside the described search space. The arbitrary initialization of the birds' locations inside the searching region is shown in Eq. (12). The candidate solution populations are represented by Eq.

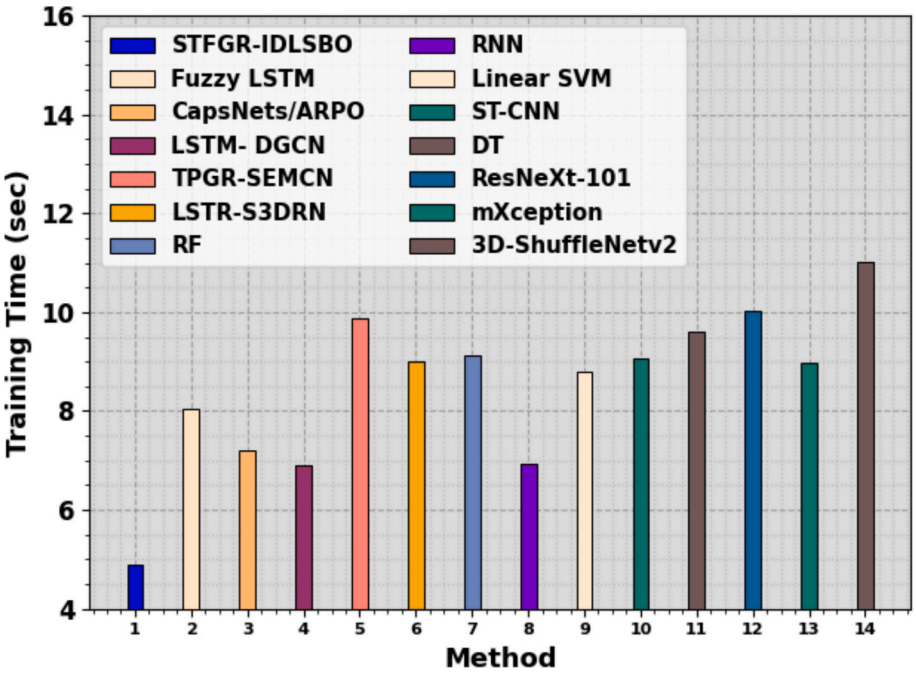


Fig. 12. TRAT analysis of STFGR-IDLSBO method with other techniques.

Table 6
Result analysis of the ablation study of STFGR-IDLSBO approach with existing methods.

Method	$Accu_y$	$Prec_n$	$Sens_y$	$Spec_y$	F_{Score}
STFGR-IDLSBO	99.25	96.28	96.24	99.58	96.22
CNN-BiLSTM	98.66	95.74	95.67	98.82	95.49
SBOA	97.88	94.98	95.01	98.03	94.76
Swin Transformer Model	97.33	94.27	94.41	97.51	94.26
ABF Technique	96.80	93.76	93.69	96.90	93.48

(13).

$$X_{i,j} = lb_j + r \times \left(ub_j - lb_j\right),\ i = 1, 2, \cdots, N,\ j = 1, 2, \cdots, Dim \tag{12}$$

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,j} & \cdots & x_{1,Dim} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,j} & \cdots & x_{2,Dim} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i,1} & x_{i,2} & \cdots & x_{i,j} & \cdots & x_{i,Dim} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{N,1} & x_{N,2} & \cdots & x_{N,j} & \cdots & x_{N,Dim} \end{bmatrix}_{N \times Dim} \tag{13}$$

Meanwhile, $X_{i,j}$ refers to the position of *ith* secretary bird and the *jth* secretary bird's question, and ub_j and lb_j indicate the particular lower and upper limitations. Furthermore, r denotes a randomly generated number, which drops amongst $(0, 1)$. N signifies the group member counts, and Dim signifies the problem variable dimensionalities. In this case, all secretary birds indicate possible solutions to address the difficulties. Therefore, the objective function is measured according to the values recommended by all secretary birds for the variable of the problems.

$$F = \begin{bmatrix} F_1 \\ \vdots \\ F_i \\ \vdots \\ F_N \end{bmatrix}_{N \times 1} = \begin{bmatrix} F(X_1) \\ \vdots \\ F(X_i) \\ \vdots \\ F(X_N) \end{bmatrix}_{N \times 1} \tag{14}$$

During this structure, F signifies the vector comprising the value of the objective function, whereas F_i characterizes the value of the objective function related to *ith* secretary bird. As the locations of the secretary birds and the value of the objective function are reviewed in all iterations, it is necessary to identify the optimal candidate solution in all cycles.

(2) Exploration stage: The secretary bird's hunting behaviour after preying on snakes is commonly divided into three different stages: looking for prey, attacking prey, and consuming prey. During SBOA, all these phases are demonstrated as shown:

Stage1: This stage uses a differential evolution model, which utilizes changes among individuals to make novel solutions, increasing global search potential or model diversity. By combining differential mutation methods, diversity helps prevent the model from becoming limited to local ideals. Therefore, the location updating for the secretary bird in the Finding Prey stage is mathematically stated utilizing Eqs. (15) and (16).

$$while t < \frac{1}{3}T, x_{i,j}^{new,P1} = x_{i,j} + (x_{random_1} - x_{random_2}) \times R_1 \tag{15}$$

$$X_i = \begin{cases} X_i^{new,P1}, \ if\ P_i^{new,P1} < F_i \\ X_i, \ else \end{cases} \tag{16}$$

Whereas t specifies the present iteration count, T signifies the total iteration counts, and $X_i^{new,P1}$ characterizes the upgraded condition of the *ith* secretary bird in the initial stage. In the meantime, x_{random_1} and x_{random_2} are candidate solutions selected randomly in this phase. Furthermore, R_1 denotes an arbitrarily produced array of dimensions 1xDim inside the interval of $[0, 1]$, with Dim demonstrating the solution space dimensionality. Finally, $x_{i,j}^{new,andP1}$ denote the value in the *jth* dimension and $F_i^{new,P1}$ designates its fitness value for the objective function.

Stage2: During this stage, Brownian motion (RB) is presented to mimic the random movements of the secretary birds. In Eq. (17), RB permits the secretary bird to sometimes break and latch upon the snake's location utilizing its bright vision. They use the theory of x_{best} combined with RB, allowing individuals to perform local hunts towards their earlier recognized optimal locations, thus improving exploration inside the adjacent solution area. This model stops individuals from early convergence to local ideals and speeds up the model's development

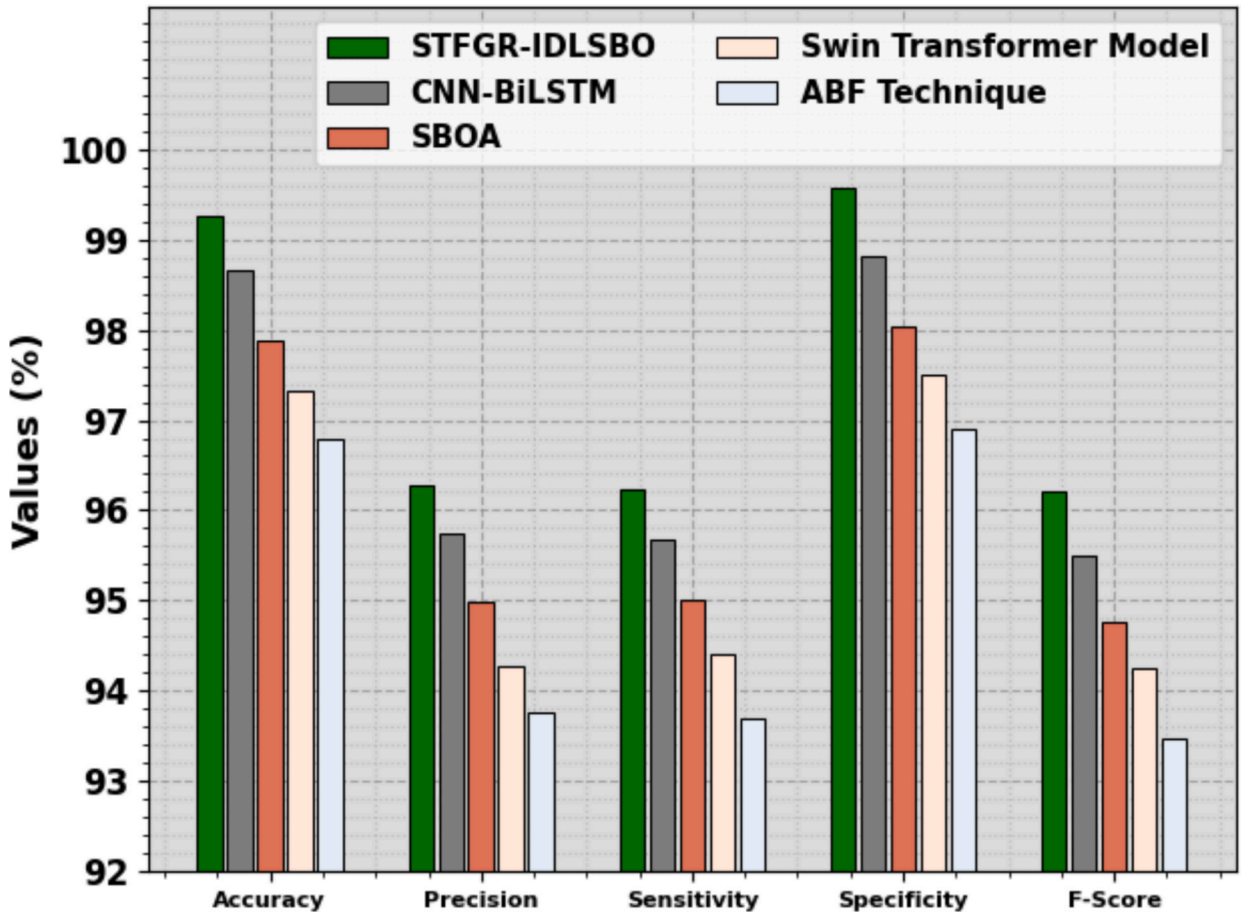


Fig. 13. Result analysis of the ablation study of STFGR-IDLSBO approach with existing methods.

towards optimum locations within the solution area. Presenting RB's randomness enhances exploration efficiency, aiding individuals in preventing local bests and attaining superior results in addressing composite issues. Therefore, the location upgrade of the secretary bird in the Eating Prey stage is mathematically modelled utilizing Eqs. (18) and (19).

$$RB = randn(1, Dim) \tag{17}$$

$$while \frac{1}{3}T < t < \frac{2}{3}T, x_{i,j}^{newP1} = x_{best} + \exp\left(\left(\frac{t}{T}\right)^4\right) \times (RB - 0.5) \times \left(x_{best} - x_{i,j}\right) \tag{18}$$

$$X_j = \begin{cases} X_i^{new,P1}, \ if \ F_i^{new,P1} < F_i \\ X_i, \ else \end{cases} \tag{19}$$

Whereas $randn(1, Dim)$ refers to a randomly generated array using dimensions 1xDim, extracted from a normal standard distribution (standard deviation of 1 and mean of 0). Temporarily, x_{best} denotes the present optimal value established.

Stage3: During this random search procedure, the Lévy fight (LF) approach supports the optimizer's global searching abilities, minimizes the probability of SBOA becoming trapped in local solutions, and improves the model's convergence precision. More prominent stages permit the model to shield the searching region more extensively, allowing individuals to attain the optimum location quicker, whereas small stages enhance the optimization accuracy.

Nonlinear perturbation features have been included to make SBOA more adaptive, flexible, and dynamic in optimization, attain a more stable exchange among exploitation and exploration, prevent initial convergence, speed up complete convergence, and improve the model's performance. Therefore, upgrading the secretary bird's location in the prey attacking stage is mathematically characterized over Eqs. (20) and (21).

$$while > \frac{2}{3}T, x_{i,j}^{new,P1} = x_{best} + \left(\left(1 - \frac{t}{T}\right)^{2\times t} T\right) \times x_{i,j} \times RL \tag{20}$$

$$X_i = \begin{cases} X_i^{new,P1}, \ if \ F_i^{new,P1} < F_i \\ X_i, \ else \end{cases} \tag{21}$$

To improve the optimization precision of the model, a weighted LF is utilized, which is characterized as RL in Eq. (22).

$$RL = 0.5 \times Levy(Dim) \tag{22}$$

(3) Exploitation stage: The secretary bird commonly uses dual major evasion tactics to protect itself or its food. Rapid flying or running is primary, and camouflage is second. The bird combines with its environment by utilizing structures or colours in the atmosphere to evade predators. Assuming these dual settings take place using similar probabilities, C_1: Blending into the surroundings; C_2: Escape by flying or running.

After the secretary bird identifies a predator close, it initially searches for an appropriate location for camouflage. If no safer and suitable camouflage surroundings are obtainable, it selects to escape by running or flying rapidly. In this case, a dynamical perturbation feature is presented, which helps the model balance exploitation and exploration. The mathematical model is shown in Eq. (23), using the upgraded states defined in Eq. (24).

$$x_{i,j}^{new,P2} = \begin{cases} C_1 : x_{best} + (2 \times RB - 1) \times \left(1 - \frac{t}{T}\right)^2 \times x_{i,j}, \; if \; r < r_i \\ C_2 : x_{i,j} + R_2 \times \left(x_{random} - K \times x_{i,j}\right), \; else \end{cases} \quad (23)$$

$$X_i = \begin{cases} X_i^{new,P2}, \; if \; P_i^{new,P2} < F_i \\ X_i, \; else \end{cases} \quad (24)$$

Here, r is allocated a value of 0.5, R_2 means an array created at random succeeding a standard distribution, x_{random} signifies the arbitrary candidate result for the present iteration, and K specifies an integer selected at random, 1 and 2. The SBOA originates a fitness function (FF) to reach the boosted performance of classification. It outlines an optimistic number to embody the better outcome of the candidate solution. In this paper, the minimization of the classification rate of error was reflected as FF. Its mathematical formulation is represented in Eq. (25).

$$fitness(x_i) = ClassifierErrorRate(x_i)$$

$$= \frac{numberofmisclassifiedsamples}{Totalnumberofsamples} \times 100 \quad (25)$$

4. Experimental validation

The performance valuation of the STFGR-IDLSBO methodology is verified using the Traffic Police Gesture dataset from the Kaggle repository [33]. The database includes 500 samples with ten classes, as defined in Table 1. Fig. 5 illustrates the sample images.

Fig. 6 exhibits the confusion matrix created by the STFGR-IDLSBO methodology under 80:20 and 70:30 of TRASE/TESSE. The outcomes indicate that the STFGR-IDLSBO approach has effectual recognition and classification of all 10 class labels.

Table 2 and Fig. 7 display the gesture detection outcome of the STFGR-IDLSBO technique. The outcomes showed that the STFGR-IDLSBO technique could find the samples proficiently. On 80 % TRASE, the STFGR-IDLSBO technique attains an average $accu_y$ of 99.25 %, $prec_n$ of 96.28 %, $sens_y$ of 96.24 %, $spec_y$ of 99.58 %, and F_{score} of 96.22 %. Besides, on 20 %TESSE, the STFGR-IDLSBO method reaches an average $accu_y$ of 99.20 %, $prec_n$ of 96.35 %, $sens_y$ of 95.50 %, $spec_y$ of 99.55 %, and F_{score} of 95.71 %.

Table 3 and Fig. 8 clearly show the gesture detection result of the STFGR-IDLSBO method. The outcomes signified that the STFGR-IDLSBO method was proficient in finding the samples. On 70 %TRASE, the STFGR-IDLSBO method achieves an average $accu_y$ of 98.34 %, $prec_n$ of 91.86 %, $sens_y$ of 91.86 %, $spec_y$ of 99.08 %, and F_{score} of 91.74 %. Besides, on 30 %TESSE, the STFGR-IDLSBO methodology gains an average $accu_y$ of 98.27 %, $prec_n$ of 91.15 %, $sens_y$ of 91.00 %, $spec_y$ of 99.04 %, and F_{score} of 90.84 %.

Fig. 9 illustrates the TRA $accu_y$ (TRAAY) and validation $accu_y$ (VLAAY) results of the STFGR-IDLSBO methodology with 80 %TRASE and 20 %TESSE. The $accu_y$ analysis is computed across the range of 0–50 epochs. The outcomes highlight that the TRAAY and VLAAY values demonstrate a rising trend, which informed the capacity of the STFGR-IDLSBO methodology and resulted in superior outcomes across multiple iterations. This is followed by the TRAAY and VLAAY leftovers closer across the epochs. These identify inferior overfitting and display maximum outcomes of the STFGR-IDLSBO method, ensuring reliable prediction on unnoticed instances.

Fig. 10 demonstrates the TRA loss (TRALO) and VLA loss (VLALO) curves of the STFGR-IDLSBO model with 80 %TRASE and 20 %TESSE. The loss values are computed within the range of 0–50 epochs. It is denoted that the TRALO and VLALO values exemplify a diminishing tendency, which informed the proficiency of the STFGR-IDLSBO model to balance a trade-off between simplification and data fitting. The constant reduction in loss values also ensures the maximum outcomes of the STFGR-IDLSBO method and tunes the predictive results over time.

Table 4 portrays the overall comparison study of the STFGR-IDLSBO approach [34–38].

Fig. 11 inspects the comparative an $accu_y$, $prec_n$, and F_{score} results of the STFGR-IDLSBO technique. The results exemplified that the Decision Tree (DT) and LSTR-S3DRN techniques displayed lesser classification outcomes. Meanwhile, the Fuzzy LSTM, Capsule Neural Networks with an augmented red panda optimizer (CapsNets/ARPO), LSTM with Deep Graph Convolutional Network (LSTM-DGCN), Random Forest (RF), RNN, Linear SVM, and ST-CNN methodologies achieve somewhat closer classification results. Simultaneously, the TPGR-SEMCN approach demonstrated reasonable performance with an $accu_y$ of 98.95 %, $prec_n$ of 92.61 %, and F_{score} of 92.67 %. However, the STFGR-IDLSBO approach shows higher performance with an $accu_y$ of 99.25 %, $prec_n$ of 96.28 %, and F_{score} of 96.22 %. Additionally, based on $sens_y$, the STFGR-IDLSBO methodology presents an improved $sens_y$ of 96.24 %, outperforming the Fuzzy LSTM, CapsNets/ARPO, LSTM-DGCN, RF, RNN, Linear SVM, ST-CN, and DT approaches, which attain lower $sens_y$ values of 92.08 %, 94.59 %, 91.96 %, 93.88 %, 93.93 %, 91.35 %, 94.05 %, and 91.20, respectively. Regarding $spec_y$, the STFGR-IDLSBO methodology depicts superior performance with a $spec_y$ of 99.58 %. At the same time, the TPGR-SEMCN, LSTR-S3DRN, Fuzzy LSTM, CapsNets/ARPO, LSTM-DGCN, RF, RNN, Linear SVM, ST-CN, and DT techniques attain lower $spec_y$ values of 94.61 %, 95.04 %, 92.61 %, 97.52 %, 92.55 %, 96.70 %, 95.08 %, 92.07 %, 97.02 %, and 91.96 %, respectively.

The training time (TRAT) outcomes of the STFGR-IDLSBO approach are related to other models in Table 5 and Fig. 12. The outcomes exemplified that the STFGR-IDLSBO approach accomplishes inferior TRAT of 4.88 s. On the other hand, the Fuzzy LSTM, CapsNets/ARPO, LSTM- DGCN, TPGR-SEMCN, LSTR-S3DRN, RF, RNN, Linear SVM, ST-CNN, and DT models obtain increased TRAT values of 8.05 s, 7.19 s, 6.90 s, 9.87 s, 9.02 s, 9.13 s, 6.94 s, 8.78 s, 9.08 s, 9.60 s, correspondingly.

Table 6 and Fig. 13 illustrate the ablation study of the STFGR-IDLSBO technique with the existing models. The study evaluates the efficiency of various techniques in improving model performance, including conventional methods, CNN-BiLSTM, SBOA, ST, and the ABF technique. The results exhibit that the STFGR-IDLSBO method outperforms the others in terms of $accu_y$ of 99.25 %, $spec_y$ of 99.58 %, and F_{score} of 96.22 %, while CNN-BiLSTM and SBOA provide competitive performance, specifically in terms of $sens_y$ and $prec_n$. The ST and ABF techniques exhibit slightly lower performance across all metrics but still maintain solid results, highlighting the relative impact of each model's architecture and approach. These findings emphasize the significance of model choice and optimization techniques in achieving the optimum balance between the diverse performance metrics.

5. Conclusion

In this paper, the STFGR-IDLSBO methodology is proposed. The main intention of the STFGR-IDLSBO methodology is to develop an efficient and robust system for gesture recognition to assist hearing-impaired persons. Initially, the proposed STFGR-IDLSBO method utilizes ABF in the image pre-processing stage to reduce noise while preserving the edges of the gestures in the captured images. Afterwards, the ST is used as a feature extractor to effectively capture multiscale representations and spatial hierarchies from gesture images. The hybrid model integrating the CNN-BiLSTM model is deployed for the gesture classification. Eventually, the SBOA will be utilized to better parameterize the CNN-BiLSTM classifier. To ensure the enhanced performance of the STFGR-IDLSBO methodology, a wide range simulation investigation is performed under the Traffic Police Gesture dataset. The performance validation of the STFGR-IDLSBO technique portrayed a superior accuracy value of 99.25 % over existing methods. The STFGR-IDLSBO technique's limitation is that it comprises a relatively small dataset, which may restrict the model's generalization capability to more diverse

gesture data. Additionally, the presented model primarily concentrates on static gesture recognition, and its performance on dynamic gestures or real-time applications remains unexamined. The study also lacks a comprehensive analysis of the model's performance in noisy or low-quality environments, which could affect its robustness in real-world scenarios. Future work could explore the integration of larger, more diverse datasets and extend the model to handle dynamic gestures and noisy conditions. Moreover, enhancing the computational efficiency for real-time applications and evaluating the system in practical settings would be valuable directions for further research.

CRediT authorship contribution statement

Mohammed S. Assiri: Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mahmoud M. Selim:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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